Joshua Prettyman, Ph.D.

I architect scalable pipelines and ML infrastructure to give data a purpose

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Personal Statement

My Ph.D. research at the National Physical Laboratory produced a novel Early Warning Signal indicator for tipping points in multidimensional dynamical systems, with applications to geospatial time series data [*ERL, 2022*]. During that time I also lectured in Mathematics and completed an internship at the Met Office.

At *Blink SEO* I built an ML-powered SaaS platform that increased productivity by 20×, taking internal tooling from spreadsheets to a marketable product: full-stack, handling everything from NLP to cloud services to UX. Since leaving Blink I have worked with many small- and medium-sized businesses to improve their data pipelines.

Technical Skills

- Regular use (10+ years) of Python: ScikitLearn, Numpy, Pandas, NLTK, Matplotlib, Plotly.
- Full stack development with Bash, Python, Google Cloud Platform (GCP), SQL, JavaScript.
- MLOps, RAG, NLP, OpenAI/Anthropic API integrations and fine-tuning, Ollama + HuggingFace.
- Data engineering (AWS, GCP, Databricks, dbt), ETL & ELT, analysis, dashboards (Tableau, Looker).
- Git, GitHub, Docker, package management, CI/CD, async, REST, GraphQL.
- Experience with C++, MATLAB, scientific computing, geospatial data, PyTorch.

Academic background

Ph.D. Mathematics — University of Reading
Thesis: *Tipping Points and Early Warning Signals with Applications to Geophysical Data*.

- Thesis develops spectral analysis methods to detect correlations in multidimensional time series, which are used as Early Warning Signals for tipping events in stochastic dynamical systems.
- Methods applied to real geospatial time series data: hurricane paths and greening/desertification.
- Publication of three papers in respected journals:
  - *A novel scaling indicator of early warning signals* (EPL, 2018)
  - *Generalized early warning signals in multivariate and gridded data* (Chaos, 2019)
  - *Power spectrum scaling as a measure of critical slowing down* (ERL, 2022)
- Implemented mathematically derived methods numerically by discretising and linearising.
- Participation in several international conferences and workshops.
- Three-month data-focussed internship at the UK Met Office (see *Professional Exp.* section).
- EPSRC funded research in industrial partnership with NPL. Attracted additional funding from NERC.

M.Res. Mathematics — Imperial College London (*Distinction*)
Dissertation: *Mesh Generation Using Solutions of an Optimal Transport Problem*.

- Dissertation uses solutions to a linearisation of the Monge-Ampère equation to generate an adaptive, optimally distributed mesh for numerical models.
- Working in C++; monitoring CPU usage and convergence time over various scenarios.
- Participation in weekly soft skills workshops and monthly internal showcases.
- Attendance at weekly Mathematics of Planet Earth seminars hosted by the Centre for Doctoral Training.
- Completed taught courses in Python, R, Dynamical Systems, Stochastic differential equations, Bayesian Analysis and Probability.

MA Mathematics — University of Edinburgh (*First Class Hons.*)
Dissertation: *Growth and its Applications in Graph Theory*.

- Dissertation explores patterns in paths through connected digraphs.
- Wrote a program with a simple UI to investigate the properties of these digraphs: this then became a teaching aid, earning a letter of thanks from the principal.
- Requisite taught courses for the Pure Mathematics degree besides optional modules in computing and mathematics education.
- Maths outreach in local schools, museum events and university seminars.

Professional Experience

Data Consultant & Architect — joshuaprettyman.com/freelance *Jan. 2025 to Present*

Delivered projects including: Model Deployment, Dashboard Creation (Tableau), Pipeline Automation (dbt, Databricks, Apache), Database and Data Warehouse Setup (AWS, GCP), Data Engineering and Data Analysis.

- I work closely with all my clients to pick apart business problems and work out plans towards solutions.
- I estimate time/cost then implement and document the agreed spec with a continuous feedback cycle.
- I maintain consistent communication with all stakeholders during planning and CI/CD phases.

Data Scientist — Blink SEO & Macaroni Software *Nov. 2021 to Dec. 2024*

Built *Macalytics*, a full-stack ML SaaS platform, from scratch. Increased team productivity by 20×, enabling year-long SEO campaigns to be delivered in weeks.

- Scaled from solo developer (me, first 18 months) to a four-person Agile team, introducing sprint-based delivery, user stories and code review as the team grew.
- Architected Python backend (*GCP Compute Engine*) to process 50M+ data daily from external services' APIs, with a *BigQuery* Data Warehouse and an integrated job-queue system (*PostgreSQL*) to run client onboarding, data imports, data analysis and ML tasks asynchronously.
- Translated workflows into automated pipelines through close collaboration and continuous deployment.
- Delivered ad-hoc analyses and data engineering for investor presentations and internal reporting.
- Developed NLP-powered keyword clustering combined with quantitative data, reducing work from days to minutes.
- Built interactive dashboards (*Plotly* + *JavaScript*) enabling clients to identify and prioritise revenue opportunities in minutes and act on recommendations directly via service integrations.
- Integrated LLMs (*huggingface*) to generate immediate-impact suggestions for content gaps.

Research Scientist — National Physical Laboratory (NPL) *Sep. 2015 to Feb. 2021*

Produced a novel scaling indicator for tipping point detection in multidimensional time series data.

- Presented results for stakeholder meetings at NPL, in published articles, and at international conferences.
- Developed methods to detect tipping points in dynamical systems, working with large geospatial time series datasets and building stochastic models to test hypotheses.
- Code written in *MATLAB* and *Python*. Data visualisations produced using *Matplotlib* and *XMGrace*.
- Accessing, cleaning, interpolating and correctly interpreting large, raw meteorological datasets.

Associate Lecturer — Sheffield Hallam University *Sep. 2017 to Jul. 2019*

Taught mathematics, statistics and computing courses from Foundation Degree to Masters.

- Taught *Microsoft Excel for Business* and *Intro to Computer Science*, besides maths and stats courses.
- Responsibilities included lesson preparation, course planning, marking, leading tutorials.
- Manned the "Maths Help" desk three times a week, providing support to students of all departments.

Informatics developer (internship) — UK Met Office Informatics Lab *Jun. to Aug. 2017*

Project assessing the viability of using amateur meteorological data to augment official observations.

- Accessed internal and external data via APIs using Java.
- Data cleaning and analyses in Python.
- Worked with the Informatics team and participated in daily stand-ups.